



Final Report:

Exploring the Potential Causes
of Leaf Yellowing in Paddy Cultivation
during the Maha Season

2022/23



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PREFACE

This report presents a comprehensive investigation into the issue of rice crop yellowing, a significant concern that emerged during the last Maha season (2022/23) in various regions of Sri Lanka.

This report introduces the reader to the context and significance of the study, emphasizing the vital role of paddy cultivation in the country's agricultural landscape. Further, the report outlines the objectives of the investigation, which aimed to unravel the reasons behind rice crop yellowing and assess its impact on farmers and rural livelihoods. The methodology employed by the team of scientists is detailed, encompassing site selection, field observations, sampling procedures, and data collection methods.

The subsequent sections of the report delve into the results and discussions that emerged from the investigation. Drone imagery and soil analyses shed light on the spatial patterns of yellowing and the variability in soil parameters across different sites. The report explores the relationships between soil nutrient levels, plant growth, and yellowing symptoms, as well as the role of root-knot nematode infections in the observed issue. The socioeconomic implications of rice yellowing on farmer income, food security, and rural livelihoods are also examined.

The report underscores the importance of farmer education and extension services in addressing the issue and promoting best agricultural practices. Trust and credibility within the agricultural extension system, impacted by evolving government policies, are highlighted as crucial elements in effectively responding to sudden agricultural challenges.

This investigation not only sheds light on the complex factors contributing to rice crop yellowing but also emphasizes the significance of scientific exploration and multidisciplinary collaboration in tackling challenges that impact the core of a nation's agriculture dependent economy.



Acknowledgment

We extend our heartfelt gratitude to the following individuals whose invaluable support and contributions greatly facilitated the successful completion of this project.

Professor Sarath S Kodithuwakku, Dean/Faculty of Agriculture, University of Peradeniya.

Ms. Nilanthi Fonseka, Assistant Director of Agriculture (Development), Department of Agriculture, Sabaragamuwa Provincial Council.

Ms. Dhananjani Bandara, Provincial Agricultural Director, Sabaragamuwa Provincial Council.

Mr. Nishantha Amarasinghe, Department of Agricultural Biology, Faculty of Agriculture, University of Peradeniya.

The team of Investigators.

The collective expertise and commitment of these individuals greatly enhanced the quality and depth of our study. Their unwavering dedication and cooperation have been instrumental in advancing our understanding of rice crop yellowing and its multifaceted implications. We are sincerely grateful for their contributions, without which this project would not have been possible.

We acknowledge the support given by Mr. Dhanushka Kumarathunga for the cover page design and Mr. W.A.K.L. Aberathna for page design and page setup of this report.

Furthermore, we express our profound gratitude for the financial support extended by the "Rukmani Kodagoda Fund," which helped to cover the expenses associated with the field visits and laboratory analyses essential for the successful execution of this project.



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Summary

Paddy cultivation plays a crucial role in Sri Lanka's agricultural landscape, providing food security and livelihoods for rural communities. In the recent Maha season (2022/23), a concerning issue of yellowing rice plants was observed in several rice-growing regions accompanied by noticeable growth retardation and yellowing symptoms. Recognizing the need for rigorous scientific investigation, the Faculty of Agriculture at the University of Peradeniya initiated an in-depth study to uncover the underlying causes of rice crop yellowing.

A specialized team of scientists focused on specific paddy tracts in the Kandy, Rathnapura, and Hambantota districts, chosen with the assistance of the Department of Agriculture. Selection criteria included crop age, proximity of affected and unaffected areas, and yellowing occurrences during sampling. Distinct tracts were identified from each district, and subsequent site visits, field observations, and farmer interviews were conducted in February – March 2023. Extensive soil and plant samples were collected, spanning both affected and non-affected areas. Soil sampling encompassed sites with confirmed yellowing and those exhibiting healthier growth, with subsequent analyses of pH, EC, and plant available P and K. Plant samples, obtained from areas both with and without yellowing, were subjected to drying and analysis for N, P, and K concentrations. The investigation into root-knot nematode infection involved the random selection of fields, calculations of infection percentages, and assessment of severity.

Measurement of soil parameters revealed substantial variability of measured soil parameters across sites. From the sites studied, 95% and 77% of sites had low contents of plant available P and K, respectively. Thus, in most of the sites, soil P and K were not at optimum levels to support crop growth. Notably, differences between sites displaying yellowing and those without such symptoms were found not statistically significant. Furthermore, soil nutrient measurements did not directly align with visual observations of plant growth and yellowing. Leaf N, P, and K concentrations largely fell within reported ranges, although specific samples indicated deficient P levels. Interestingly, a positive correlation was observed between leaf N and K concentrations. Importantly, the occurrence and severity of yellowing symptoms did not solely correspond to P and K deficiencies.

Based on % infected plants and galling index, the incidence and the severity of root knot nematode infection between the two types of fields (i.e. yellowing symptoms and healthy looking) did not differ significantly ($P= 0.05$). Root knot nematode infections at varying incidence (6-83 %) were reported from rice and weed plants of all three locations with % occurrence ranging from 14- 75. However, the severity of the nematode infection both in rice plants and weed plants was not severe, being 2 on the maximum galling scale of 10. A significant negative correlation (-0.63089) was identified between soil phosphorous content and galling index in rice roots ($P< 0.0278$). Significant positive correlations were demonstrated between % nematode incidence of rice and galling index of rice, galling index of weeds and % nematode incidence in weeds. In all locations, nematode infection was



identified as a potential threat to the crop grown in next seasons, especially when P and K nutrients are not sufficiently available.

Considering the socioeconomic implications, the impact of paddy yellowing had adverse effects on farmer income and food security. The trust and credibility of extension officers eroded due to evolving government policies on agrochemical applications were recognized, as needing rebuilding through field demonstrations that promote best agricultural practices.

In conclusion, despite the lack of significant differences in soil parameters between areas displaying rice yellowing and those without, a potential link between sub-optimal nutrient inputs and crop performance emerged. Results indicated that there is a high risk of spreading the nematode infection to the next crop. Consequently, implementing measures for managing the root-knot nematode infection in rice cultivation is crucial. The study highlighted the pivotal role of nutrient management practices and soil nutrient availability in influencing crop health and overall performance in the studied paddy fields.

As a recommendation, the study underscores the necessity of prioritizing nutrient management practices, refining fertilizer application strategies, augmenting soil nutrient availability and reducing the spread of nematode infestation, to address rice yellowing issues and enhance overall crop productivity. Further research, with a more comprehensive dataset, is crucial to establishing a more robust understanding and confirmation of the relationship between nutrient inputs and rice crop health.

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Introduction

Paddy cultivation holds immense significance in Sri Lanka's agriculture sector, ensuring food security and employment opportunities for a substantial rural population. The cultivation of paddy takes place in two main seasons, namely "Yala" and "Maha," corresponding to the monsoonal rains. While paddy is cultivated in all districts of Sri Lanka, the main paddy-producing regions include Polonnaruwa, Ampara, Kurunegala, Anuradhapura, and Hambantota.

During the last Maha season (2022/23), an alarming occurrence of yellowing rice plants during the vegetative stage was reported in various paddy cultivating areas, including Hambantota, Rathnapura, and Kandy. This condition was characterized by prominent yellowing and discoloration of rice plants, along with growth retardation, and was reported by both farmers and field-level extension personnel in multiple districts.

Upon primary investigations, the Rice Research and Development Institute (RRDI) identified several potential causes of the yellowing disease. Notably, nutrient deficiency emerged as a primary reason, stemming from the reduced rate of application of chemical fertilizers such as urea, triple super phosphate (TSP) and muriate of potash (MOP) over previous cultivation seasons. This deficiency has led to a scarcity of essential nutrients nitrogen, phosphorus, and potassium in the soil, subsequently causing the rice plants to exhibit yellowing symptoms. Additionally, nematode infestations were suspected to be another major contributor to the yellowing disease. Root-knot nematodes inflicted damage to the root system, impeding the uptake of nutrients, thereby exacerbating the nutrient deficiency in the plants. Furthermore, adverse environmental conditions, such as airborne dust particles originating from India and extreme cold weather triggered by a cyclone, speculated to be further deepened the issue, leading to widespread leaf yellowing in crops planted after 15th November 2022 (<https://www.sundaytimes.lk/230122/news/rice-plant-diseases-likely-to-destroy-10-percent-of-maha-harvest-509091.html>).

To combat these challenges and safeguard the rice harvest, the RRDI proposed a series of potential remedial measures. One of the measures was to implement balanced fertilizer application, adhering to the recommended ratio of 70 percent chemical fertilizers and 30 percent organic fertilizers. Additionally, farmers were requested to adopt nematode control measures, including the use of nematicides and crop rotation practices, to mitigate nematode infestations. Timely reporting of disease outbreaks, coupled with agricultural extension services, was identified crucial in providing prompt guidance and support to affected farmers. Furthermore, importance of continuous research and development efforts, alongside weather monitoring and adaptation strategies, was further highlighted in effectively tackling the yellowing disease and ensuring sustainable rice cultivation in Sri Lanka (<https://www.sundaytimes.lk/230122/news/rice-plant-diseases-likely-to-destroy-10-percent-of-maha-harvest-509091.htm>).

As the pioneer in agriculture higher education in Sri Lanka, Faculty of Agriculture of University of Peradeniya recognized the complexity in identifying the underlying causes of leaf yellowing in plants without thorough in depth scientific exploration. To address this issue, a dedicated team of scientists from the Faculty of Agriculture of the University of Peradeniya, representing various related academic disciplines, embarked on a comprehensive investigation into the issue of yellowing in the rice crop. The primary objectives of this study were to delve into the reasons behind the yellowing of rice crops, assess its impacts on farmers, and propose effective strategies to mitigate its occurrence.



- Collecting samples from a field with three months old poorly grown paddy crop in Weeraketiya, Hambantota district.



- A field with slightly yellowed and stunted rice crop (variety - H4) in Ankumbura, Kandy district.

Methodology

The team of scientists identified paddy tracts to be visited in Kandy, Rathnapura and Hambantota districts with the support of the officers of the Provincial Departments of Agriculture and the Department of Agriculture. In selecting the paddy tracts, the age of the crop, the presence of affected as well as non-affected areas in close proximity in the same tract and the presence of yellowing incidences at the time of sampling were considered. Accordingly, one paddy tract each from Ankumbura, Nivithigala and Weeraketiya was identified from Kandy, Rathnapura and Hambantota Districts, respectively. Site visits and sampling were conducted from 9th to 22nd February, 2023. In the Rathnapura district, samples were collected from an additional tract in the Pelmadulla region. At the time of sampling the crop was at the late vegetative or early reproductive stage.

In each site, incidences of yellowing of the rice crop were recorded based on field observations and information gathered from the farmers and government officers. Observations on nematode infestation in plant roots were made following standard procedures as described later. Soil and plant samples were collected from the selected fields. Non-affected areas or fields with relatively good plant growth were included in recording observations and sampling whenever available. Farmers and government officers were also interviewed during the site visits.



- Research team collecting soil and plant samples, inspecting plant roots for nematode infections and exploring the region using drone assisted imaging.

Soil sampling and analyses

Soil samples were collected from paddy fields that had incidents of yellowing at the time of sampling or as reported by the Agriculture Instructors. Areas that had relatively low or no symptoms of yellowing were also included in the sampling when available in the regions visited. Accordingly, the number of sites selected for soil sampling in the paddy tracts from Ankumbura, Weeraketiya and Rathnapura regions were 8, 11 and 7, respectively. Altogether there were 17 sites with confirmed yellowing incidences and 9 sites with relatively better plant growth with low or no incidences of yellowing. The number of sites common for plant sampling, soil sampling and observing nematode incidences in Ankumbura, Weeraketiya and Rathnapura were 6, 7 and 7 sites, respectively.

Soil samples were collected from three locations in each site at a depth of 0-10 cm and composited to obtain a representative sample for the site. However, it should be noted that some farmers had applied organic manure in large quantities during the previous season and also had applied some synthetic fertilizers to their paddy fields. The exact dates of such applications with respect to the soil sampling date could not be revealed during the field visits. Collected soil samples were transported to the Department of Soil Science, Faculty of Agriculture on the same day of sampling. Soils were air-dried and passed through a 2-mm sieve before using in analyses. All soil analyses were performed on three laboratory replicates of each soil sample. Soil pH was measured in a soil: water suspension of 1:2.5 using a pH meter (EUTECH CON500). Soil EC was measured in the soil: water suspension of 1:5 using an EC meter (EUTECH 510). Mehlich-3 extractable potassium (K) and phosphorous (P) in soils were determined using spectrophotometry following standard protocols (Mehlich 1984; Sikora and Moore 2014).



• Collecting soil samples from a paddy field in Nivithigala, Rathnapura district.

Leaf N, P and K analysis

Plant samples were collected randomly from the fields selected to obtain soil and nematode observations. This included areas (patches) with and without yellowing symptoms in rice crops. From those selected locations, at least five rice hills were uprooted, and the roots were separated. Shoots were taken to the laboratory, dried in an oven at 60 °C for two days and ground before element analysis. Leaf N concentration was determined using the Kjeldahl method (Nelson and Sommers 1973). For the analysis of leaf K concentration, samples were digested in nitric acid and tested using a flame atomic absorption spectrophotometer (GBC model 932AA) (Van Ranst et al. 1999). For the determination of leaf P concentrations, approximately 100 mg subsample was digested in nitric/perchloric acid (400 °C for 3 h) and analyzed using the molybdo-vanadophosphate method (Kitson and Mellon 1944).

Investigation on root knot nematode infection

The fields were selected randomly within a tract representing well-grown and poorly-grown crops. The number of fields subjected to the investigation was; 17 from Ankumbura, 12 from Weeraketiya and 14 from Ratnapura.

Data collection and calculations:

1. Percentage area showing yellowing in a given field: This was done based on the visual observations, and the area showing yellowing per field was estimated.
2. Percentage occurrence of root-knot nematode infection in rice and weed plants: This was calculated separately for each location using the following equation.

$$\frac{\text{No. of fields reported root-knot nematode infections}}{\text{Total no. of fields used for observation}} \times 100$$

Note: When the occurrence of nematode infection in healthy-looking or poor-looking fields calculated, the total number of such fields used for observation was used for the calculation.

3. Incidence of root-knot nematode infection in rice plants: Randomly selected rice plants (representing both yellowing/poorly grown and well grown) were uprooted and the root system was observed for the presence of characteristic hook-shaped galls at the root tips. This is a reliable symptom to confirm the root-knot nematode infection. Accordingly, the percentage root-knot nematode infection in rice plants was calculated as follows

$$\frac{\text{No. of plants showing root knots}}{\text{Total no. of rice plants uprooted for observation}} \times 100$$

4. Severity of root-knot nematode infection: Root systems were ranked using the galling index chart (Bridge et al., 2005). The scale ranged from 0-10.

5. Incidence of root-knot nematode infection in weed plants: Randomly selected weed plants (representing monocotyledonous and dicotyledonous, but depending on the availability) were uprooted and the root system was observed for the presence of characteristic hook-shaped galls at the root tips. Accordingly, the percentage root-knot nematode infection in rice plants was calculated as follows.

$$\frac{\text{No. of plants showing root knots}}{\text{Total no. of rice plants uprooted for observation}} \times 100$$

6. Severity of the root-knot nematode infection: The severity of infection was ranked based on the galling index chart (Bridge et al. 2005).

Data Analysis

To test the difference between sites with and without leaf yellowing symptoms for the variables studied, the Wilcoxon rank sum test was performed. This included the variables such as; soil- pH, EC, P and K concentrations, plant tissue- N, P and K concentrations, and the intensity of yellowing symptoms developed in the leaves (as a percentage). The strengths of the linear relationship between the above variables were tested using Spearman rank correlation. All the interpretations were made at $\alpha=0.05$. Data were statistically analyzed using SAS statistical software.



- Inspecting signs of nematode infection in weeds collected from paddy fields

- Root system of a weed plant showing root galls due to nematode infection



Results & Discussion

Soil variability across sites

Drone images captured in Ankumbura and Weeraketiya sites are provided in the Figure 1A and 1B respectively. Reflectance maps (on yellow and red color scale) have highlighted less greener areas by yellow to red color. At the time of sampling, the yellowing patterns in these fields were highly scattered and mixed with natural yellowing observed in the end of the growing season in some fields in the tract.

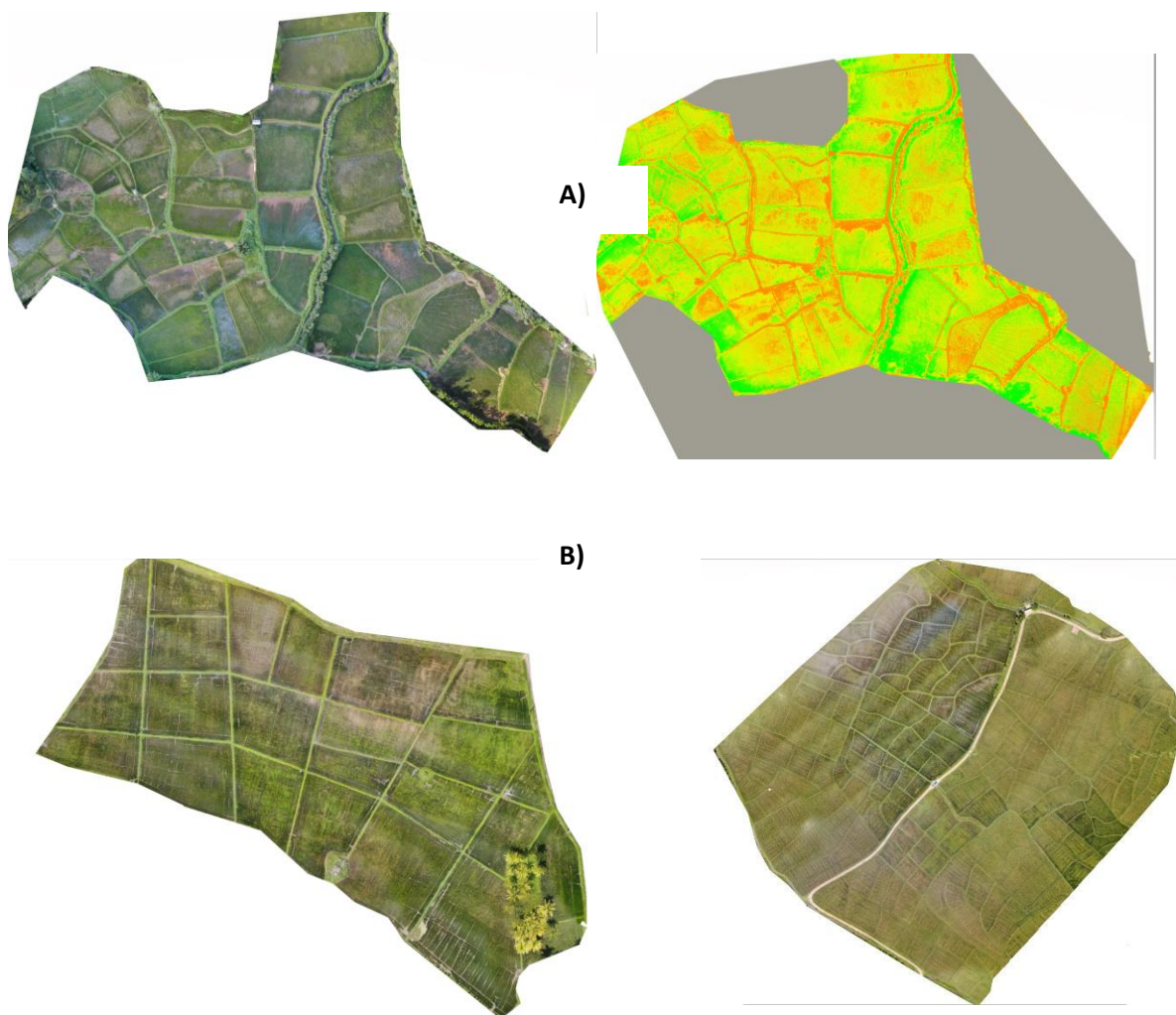


Figure 1. A) Drone images of Ankumbura (left RGB reflection and right reflectance) and B) Weeraketiya sites.

The measured soil parameters indicated high variability across sites (Figure 2. A to 2.D and Table 1). Soil pH ranged from 4.96 to 8.13 and EC ranged from 6 to 220 $\mu\text{S}/\text{cm}$ (Table 1). Plant available P and K levels in soils as determined through the Mehlich-3 extraction method ranged from 0.40 to 32 mg/kg and 20 to 146 mg/kg, respectively. In general, according to soil fertility indices, soil P and K levels are considered as low when the Mehlich-3 extracted P and K levels are below 20 and 90 mg/kg, respectively (Sikora and Moore, 2014). Accordingly, from the 22 sites, 95% and 77% of sites had soils low in plant available P and K contents.

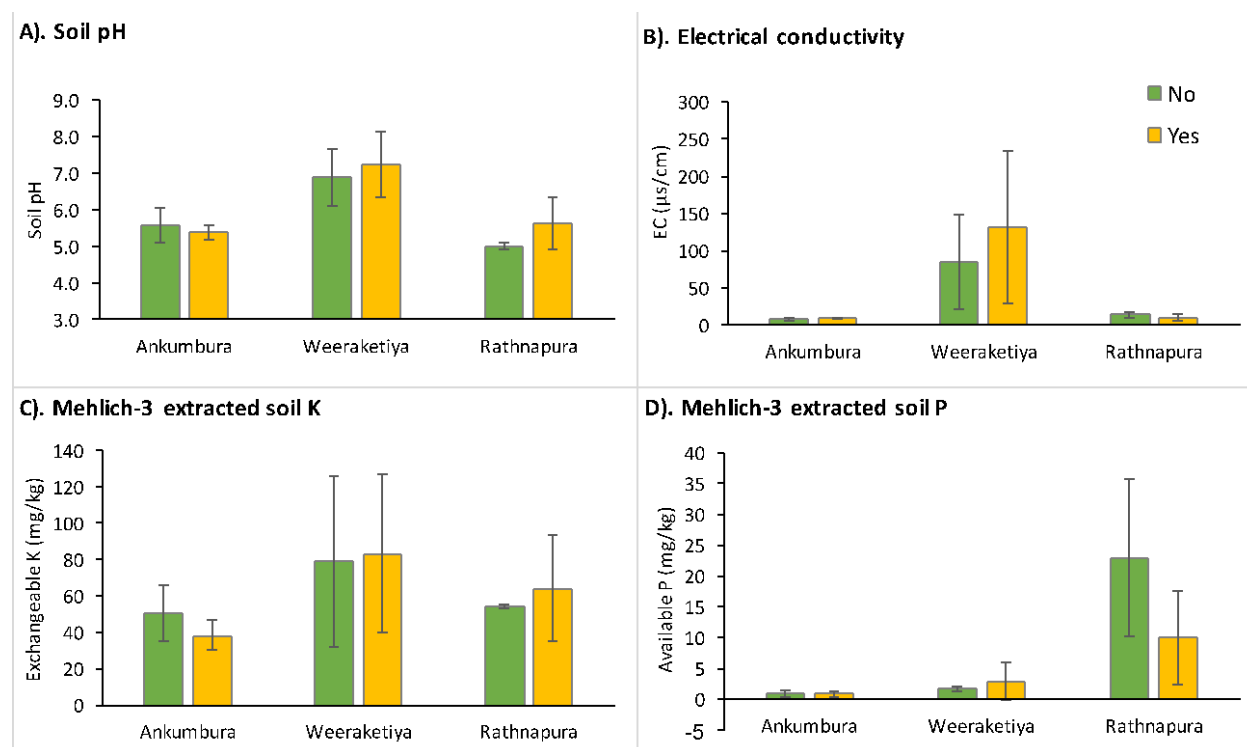


Figure 2. Variability of the soil parameters measured in soils collected from sites with and without incidences of yellowing (Yes – yellow and No – Green, respectively) in selected paddy tracts in the three regions; Ankumbura, Weeraketiya and Rathnapura.

Soil nutrient status & yellowing

In overall, the measured soil parameters were not significantly different ($P>0.05$) between sites with and without yellowing incidences (Table 2). In Sri Lanka, the commonly used methods for determining plant available P and K contents in soils are Olsen-P (OP) and Ammonium Acetate extractable-K (AA-K) methods. Mehlich-3 extracted P level of 20 mg/kg corresponds to about 10 mg/kg of P when determined using OP method; whereas, Mehlich-3 extracted K level of 90 mg/kg corresponds to about 79 mg/kg of K when determined with AA-K method. All the sites sampled in Ankumbura and Weeraketiya paddy tracts and three sites in the Rathnapura paddy tract had very low available P contents because the values were below 10 mg/kg (OP equivalent of 5 mg/kg). It should be noted that soil acidity is relatively high in sites from Ankumbura and Rathnapura and these areas are also prone to iron toxicity because of the soil types and agroecology in the regions.

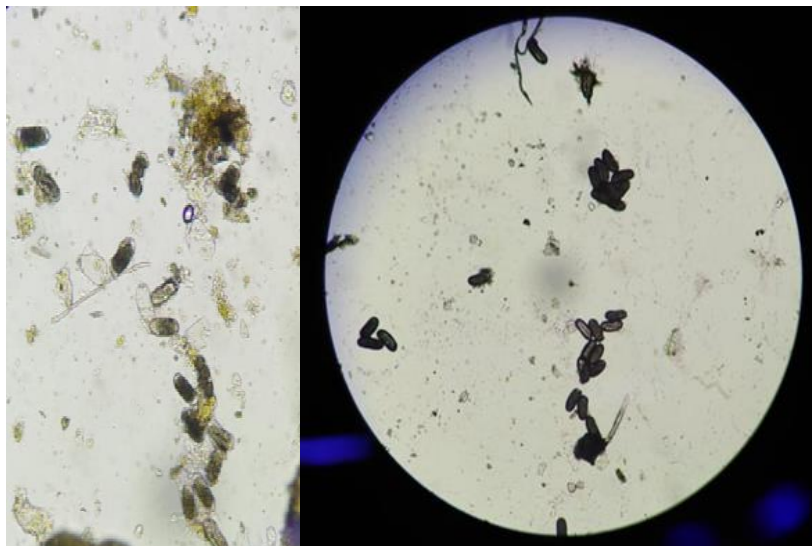
Rust in standing water was observed in some of these sites. Soil pH, EC and plant available K contents in soils were relatively higher in soils collected from Weeraketiya than the other two sites (Figure 2. A, Figure 2. B and Figure 2. C). Overall, significant positive relationships were observed between EC and plant available K content ($r^2 = 0.50$, $P = 0.024$) and EC and soil pH ($r^2 = 0.63$, $P = 0.003$). It should be noted that soil nutrient levels measured at one-time point will not provide a true picture of the nutrient availability for the crop throughout the cropping season. Nutrient level in the soil at a given time is decided by management practices, numerous nutrient transformations happen in soil and plant uptake. High levels of K in some fields could be due to the application of MOP prior to the collection of soil samples. However, the results confirmed that in Ankumbura and Weeraketiya, most of the site's soil P and K levels are not at optimum levels to support crop growth.

In all three locations, it was noted that fertilizer application was not followed according to the recommendations due to the unavailability of the materials on time. Triple super phosphate (TSP) was applied by only four farmers at or below the recommended dosage (Weeraketiya - three and Rathnapura - one). Urea and MOP were not available on time and at the required rate for the farmers in the regions. All the sites in Weeraketiya and Rathnapura that showed relatively good plant growth and less or no incidences of yellowing had received fertilizers close to the recommended dosage and at the correct time or located in the lower part of the tract where water flowing from other fields are usually gathered. In addition, in one site with good plant growth in Rathnapura, the farmer applied biochar (pyrolyzed paddy husk) and practiced transplanting of rice and rotary weeding. In this site, the amount of fertilizers applied was below the recommended level but the above mentioned agronomic practices may have helped to ameliorate the soil conditions to support plant growth. All paddy fields that showed yellowing symptoms and/or poor plant growth had received less quantity of fertilizers than the recommended levels, at incorrect proportions, and not at the right times.

According to the information gathered from the farmers and the Agriculture Instructors, most of the sites (7 out of 11) in the paddy tract in Weeraketiya received comparatively high levels

of compost (at or above 5 t/ha) and liquid organic fertilizers during the previous and the present season. In these sites, the EC was higher than in the other sites in the paddy tract that received low or no compost. In some sites, plants looked like those affected by salinity and had very poor growth. Soil EC and plant available K levels did not have a significant correlation in Weeraketiya. Therefore, it will be worthwhile to test the presence of other ionic species in these soils such as Ca^{2+} , Na^+ , Mg^{2+} , Cl^- and SO_4^{2-} to investigate the reason for increased EC. In Rathnapura, farmers who had received compost under the Government subsidy scheme, had applied them at a rate less than 0.5 t/ha. Except in two sites, in all other sites the farmers applied compost after establishing the crop during seedling stage.

In Ankumbura, relatively good plant growth was observed in a few patches in paddy fields and those were considered sites with relatively good growth of plants. These distinct areas were the places where paddy straw has been piled in previous seasons as gathered from the discussions with the farmers. However, plant available nutrient levels were not significantly different in those locations from surrounding areas that had yellowing incidences and/or poor plant growth.



- Microscopic view of crushed root galls showing, immature eggs and stage 1 juveniles of the root knot nematodes

- A three months old poorly established rice crop in Ankumbura, Kandy district. The crop could not withstand recent heavy rains due to poor growth.



Plants related findings.

Soil nutrient measurements did not provide sufficient evidence to support the visual observations made on plant growth and yellowing. On the other hand, visual observations could not reveal the actual nutrient status of the plants due to “hidden hunger”. Therefore, a plant that did not show any yellowing symptoms could still be suffering from nutrient deficiencies. When comparing leaf N, P and K concentrations, the mean concentrations were within the levels reported in most of the previous studies (Mayamulla et al. 2017; Suriyagoda 2022). However, the values were within a broad range, and therefore, some of the plant samples had deficient levels of P concentrations in their green leaves. As a benchmark, straw N, P, and K concentrations less than 1%, 0.06% and 0.18%, respectively at maturity are generally considered deficiency thresholds for N, P, and K when identifying soils, which have a low capacity to supply N, P and K to rice plants (Dobermann *et al.*, 1998; Dobermann & Fairhurst, 2000; Vandamme *et al.*, 2016).

Leaf N and K concentrations were positively correlated ($r^2 = 0.46$, $P = 0.016$) (Table 2). Moreover, leaf P concentration was positively correlated with leaf K ($r^2 = 0.44$, $P = 0.021$), and negatively correlated with soil pH ($r^2 = -0.68$, $P = 0.001$), and EC ($r^2 = -0.71$, $P = 0.005$).

Table 1. Descriptive statistics of the variables studied

Variable	N	Mean	Std dev	Median	Minimum	Maximum
Leaf N (%)	27	2.98	0.9	3.0	1.6	4.9
Leaf P (%)	27	0.13	0.1	0.1	0.01	0.3
Leaf K (%)	27	0.97	0.5	1.0	0.2	2.0
Soil pH	20	6.01	1.0	5.6	5.0	8.1
Soil EC	20	47.7	69.9	11.7	6.0	220.4
Soil P*	20	2.8	4.2	1.0	0.2	16.0
Soil K*	20	52.1	27.3	43.0	17.8	128.4
Nematode_rice	26	21.2	31.6	0.0	0.0	100.0
Galling_rice	18	0.7	0.7	1.0	0.0	2.0
Nematode_weeds	21	33.3	45.9	0.0	0.0	100.0
Galling_weeds	19	0.6	0.9	0.0	0.0	2.0

Note: Nematode_rice; % occurrence of root-knot nematode infection in rice, Galling_rice; Severity of root-knot nematode infection in rice, Nematode_weeds; % occurrence of root-knot nematode infection in weeds, Galling_weeds; Severity of root-knot nematode infection in weeds.

*The values reported for P and K are equivalent for the Olsen P and Ammonium Acetate extractable K, respectively.

Leaf N concentration was higher in plants without showing yellowing symptoms than in those with yellowing symptoms (Table 2). However, leaf P and K concentrations were similar between the two groups. Tissue P and K data suggest that the yellowing is not due to the limitation of P and K, although P and K concentrations in some of the plant samples were very low.

Table 2: Effects of leaf yellowing on other variables studied

Variable	Yellowing	N	Mean	Stderr	P> t
leaf [N]	No	8	3.54	0.24	0.0214
	Yes	12	2.84	0.17	
leaf [P]	No	8	0.16	0.04	0.6759
	Yes	12	0.14	0.03	
leaf [K]	No	8	1.27	0.14	0.1768
	Yes	12	0.96	0.15	
Soil pH	No	8	5.95	0.36	0.8368
	Yes	12	6.05	0.30	
Soil EC ($\mu\text{s}/\text{cm}$)	No	8	46.52	20.55	0.9514
	Yes	12	48.55	22.90	
Soil P* (mg/kg)	No	8	3.39	1.97	0.6366
	Yes	12	2.45	0.93	
Soil K* (mg/kg)	No	8	48.07	6.49	0.6034
	Yes	12	54.79	9.37	
Nematode rice	No	8	14.23	8.76	0.7859
	Yes	12	17.64	8.21	
Galling rice <i>Index</i>	No	5	0.80	0.37	0.8993
	Yes	7	0.86	0.26	
Yellowing	No	8	3.75	3.75	0.2547
	Yes	7	17.14	11.28	
Nematode_weeds	No	7	28.51	15.30	0.8675
	Yes	12	25.00	13.06	
Galling weeds	No	7	0.71	0.36	0.3082
	Yes	10	0.30	0.21	

*The values reported for P and K are equivalent for the Olsen P and Ammonium Acetate extractable K, respectively.

The status of root-knot nematode infections

The % root knot nematode infected rice plants, galling index (a measure to express the severity of the nematode infection) of rice, % weed plants in the same paddy fields showing nematode infections and the galling index of weeds did not differ significantly between the fields having healthy-looking plants and the fields showing yellowing symptoms (Table 2). Therefore, it is evident that the incidence and the severity of root knot nematode infection between the two types of fields (i.e. yellowing symptoms and healthy looking) are similar.

The occurrence of root knot nematodes (RKN) in all locations was evident in both rice and weed plants with varying percentages (i.e. 14-75) (Table 3). Results also revealed the occurrence of root knot nematode infection in the fields having a healthy appearance (no yellowing) and with the better shoot and root growth and it ranged from 29-100 % in rice plants. Weed plants collected from the fields demonstrating better growth also harboured root knot nematodes with a % occurrence ranging from 25-60, indicating the presence of nematodes even in the fields with a healthy appearance (Table 3).



- Inspecting fields in Weeraketiya, Hambantota District with the support of officers from the Provincial Department of Agriculture and the farmers.

Table 3: Summary of root-knot nematode infections in the studied fields

Parameter	Ankumbura		Weeraketiya		Ratnapura	
	In rice	In weeds	In rice	In weeds	In rice	In weeds
% occurrence of Root knot nematodes (with respect to the total number of fields)	18.8	14.3	64.2	14.3	42.8	75.0
% occurrence of root knot nematodes in the fields having healthy appearance	100	33.3	28.4	25.	75.0	60.0
% occurrence of root knot nematodes in the fields showing yellowing and poor growth	0	0	35.7	25.0	57.1	28.5
% incidence of root knot nematodes	21.0	6.3	26.95	83.0	24.0	61.0
Severity (galling index)	1-2 (10-20% of the root system had root galls)	2 (20% of the root system had root galls)	1-2 (10-20% of the root system had root galls)	1 (10% of the root system had root galls)	1-2 (10-20% of the root system had root galls)	1-2 (10-20% of the root system had root galls)
Weeds reported root knot nematode infections		Bajiri		Bajiri, Thunessa		Bajiri, Thunessa, Wal karabu Weedy rice

The percentage incidence of RKN infection in rice plants and weed plants ranged from 6 to 83 in the three locations, demonstrating the level of infection present in the fields at the time of observation. However, the severity of the nematode infection both in rice plants and weed plants was not severe, being 2 on the maximum galling scale of 10.

It means that only 20% of the root systems have been affected by the gall formation, hence 80% of the root system is in a functional state for water and nutrient uptake and also for anchoring purposes. Further, there were fields in Ankumbura, which reported no RKN infection though they exhibited yellowing and poor plant growth (Table 3). Results revealed the common weed hosts that harbour root knot nematodes, hence the importance of controlling weeds acting as alternative hosts in the rice fields.

Table 4 summarizes the significant negative correlations identified between different aspects and parameters related to nematode infection.

Table 4: Significant negative correlations identified in the study

Associations tested	Correlation coefficient, probability and sample size
1. Leaf Phosphorous content and % area of yellowing in the field	-0.49996, 0.0346, 18
2. Soil Phosphorous content and galling index in rice roots	-0.63089, 0.0278, 12

The negative relationship between the galling index of rice and soil P (i.e. high galling under low Phosphorous soils) observed in the present study (Table 4), agrees with the explanation by Pe' ret et al. (2011). According to Pe' ret et al. (2011), root system architecture (RSA) is greatly altered upon phosphate depletion, resulting in a shallower root system bearing more and longer lateral roots as well as denser and longer root hairs and these modifications in the root system are favourable to root knot nematode infections. As nematodes are aerobic and prefer to be at shallow depths shallower roots have a higher possibility to get higher RKN infection than the deeper roots under anaerobic conditions. Further, denser root hairs provide a higher number of entry points to the nematodes, hence, the number of galls could be high and it will result in a higher galling index. Further, Rose et al. (2013), have demonstrated the higher root hair density and root length in rice cultivar "Nipponbar" under Phosphorous deficient soils.

Table 5 summarizes the significant positive correlations identified between different aspects and parameters related to nematode infection.

Table 5: Significant positive correlations identified in the study

Associations tested	Correlation coefficient, probability and sample size, respectively
1. Soil pH and galling index in rice roots	0.63583, 0.0263, 12
2. Soil Phosphorous and % nematode incidence in weeds	0.46558, 0.0445, 19
3. Soil K and % nematode incidence in weeds	0.50318, 0.0281, 19
4. % nematode incidence in rice and galling index of rice	0.82789, <0.0001, 18
5. % nematode incidence in rice and % nematode incidence in weeds	0.44001, 0.045, 21
6. % nematode incidence in rice and galling index in weeds	0.47769, 0.0386, 19
7. % nematode incidence in weeds and galling index in weeds	0.96581, <0.0001, 19

We observed a positive relationship between the soil pH and the galling index in rice roots (Table 5), where the galling index is an indicator of the number of successful infections by the RKNs. For successful infection, aggregation of the infectious stage of RKNs is a prerequisite. Usually, the nematode population shows a negative correlation with soil pH (Nishantha et al., 2020). Wang et al. (2009) have observed higher aggregation of several RKNs at a range of pH of 4.5 – 5.4. However, the preferred pH for the level of aggregation depends on the RKN species/strain. Soil pH in our study reported a range of 5.0 – 8.1 with a median of 5.6, hence a higher level of aggregation of the infectious stage of RKN would be a possibility, and it could have resulted in a higher galling index in rice roots. Soil pH and soil EC have a positive relationship with the % area of the yellowing in the field. Higher soil pH and EC (i.e. Salinity stress), can cause imbalances in nutrient uptake, hence leaf discolorations are possible.

Potassium provides resistance to rice plants against biotic stresses including the RKN infection (Liu et al., 2022). As the rice roots show resistance to RKN infection, weed plants would be preferred by the nematodes. Rice root-knot nematodes have a wide range of host plants including monocots and dicots, grown in paddy fields (Bridge et al., 2005)



A positive relationship between RKN incidence and the galling index both in rice and weeds indicates the successful infections caused by the nematodes in those host plants. It also highlights the susceptibility of the rice varieties grown in the study area and the weeds in the rice fields to RKN and the conducive conditions provided by the plants for the reproduction of the nematodes.

Therefore, the selection of resistant rice varieties for RKNs and the removal of favourable weed plants can be considered effective management measures against rice root knot nematodes.

Socio-economic aspects of paddy yellowing

Farming is the main livelihood of farmers who were interviewed from Hambantota (Weeraketiya). That was different when it comes to Ratnapura (Nivithigala and Pelmadulla) and Kandy (Ankumbura). Farmers in Rathnapura and Ankumbura cultivate paddy mainly for their food security rather than the main source of livelihood. According to the key informant discussions, and the interviews of the farmers; it was revealed that most farmers in Nivithigala are tea smallholders and that is their primary source of agricultural income. In addition, some of them engaged in gem mining as well, mostly as laborers.

The yellowing of rice crop and or the growth retarded paddy fields were observed in all three villages in three districts studied at different levels. However, the farmers' understanding and knowledge about the situation were very low. According to their perception, not having an adequate chemical fertilizer for their paddy fields due to banning imposed on chemical fertilizer and high price is the only causative factor for paddy yellowing. They were not knowledgeable about the other possible causative factors identified as the imbalance application of fertilizer, pests and diseases, soil conditions such as iron toxicity or salinity.

Most of the Farmers who participated in the interviews claimed that the growth of their rice plants have retarded and not expecting the yields they obtained in previous seasons. Reduced crop yield directly affects the income and livelihoods of farmers who heavily rely on paddy cultivation. Moreover, the reduced yield directly affects food security. Economic losses due to the yellowing of the crop will result in the indebtedness of the farmers, especially those who directly rely on farming as their livelihood.

It was not observed a direct relationship of demographic factors such as age, gender, income and education with the yellowing of paddy.



Farmer Education and Extension Services

According to the observations and data collected, paddy yellowing poses a significant challenge to paddy farmers in studied locations, affecting crop productivity, farmer income, and food security. However, it was observed that not all the paddy fields in the area were similarly affected by yellowing. In the conversations with such farmers, it was revealed that farmers who had relatively good crop stands used a balanced fertilizer and followed good agricultural practices comparatively to the farmers who owned paddy fields affected by yellowing. Therefore, enhancing farmers' knowledge through training programs and extension services is crucial for promoting awareness about paddy yellowing causes, symptoms, and appropriate management practices. It is needed to take immediate mitigation efforts.

The overall best management practices and Good Agricultural Practices (GAP) including balanced fertilizer application, efficient water management, and maintaining optimal soil health, should be promoted through agricultural extension. The need for immediate responsiveness of the agricultural extension system in a sudden situation like rice crop yellowing was identified. In general, the farmers were not satisfied with the timely responsiveness of the agricultural extension and research system. According to the key informant discussions with the agricultural instructors and the regional management level officers, it had taken a fairly long time to give a fertilizer subsidy for the affected areas due to bureaucratic administrative procedures of the government.

Another challenge faced by the field level extension officers was the loss of trust, accountability and credibility as the field level extension officers, due to changing policies of the government on agrochemical applications including fertilizer. To regain the lost trust and credibility, it is recommended to conduct field demonstrations applying good agricultural practices in nutrient, pest and disease management.



- Inspecting fields in Ankumbura, Kandy district with the support from the officers of Department of Agriculture and farmers.



- Collecting information through informal discussions with government officers and farmers

Conclusions & Recommendations

The plant available P and K in most of the studied soils were well below the critical levels at the time of sampling. Although there was no statistically significant difference in soil parameters between the areas with and without rice yellowing incidences, crop performance was different between the two groups. Lack of relationship between rice yellowing incidences and soil nutrient levels could be due to sub-optimal nutrient inputs to the sites and the differences in plant growth could have masked nutrient deficiency symptoms and crop performance. High variability of measured soil parameters and plant growth, naturally due to different agro-management practices adopted by farmers, did not allow us to test the link between nutrient management and yellowing and arrive at a definitive conclusion. However, sub-optimal nutrient application to paddy fields resulting in low nutrient availability seems to be one of the most important factors that determined crop performance in the studied paddy fields. In support of this, we observed that leaf N concentration was higher in plants without showing yellowing symptoms than in those with yellowing symptoms although leaf P and K concentrations were similar between the two groups. Further, successful nematode infestation as indicated by galling incidences in rice roots increased with decreasing soil P availability. Therefore, maintaining plant available nutrient contents at optimum level is crucial for a healthy crop.

Root-knot nematode infection was observed in both fields displaying yellowing symptoms and fields with seemingly healthy plants. A strong positive correlation between root-knot nematode incidence and galling index in both rice and weed plants indicates that nematodes can successfully infect both rice and weed plants. It is important to note that the weed plants present in the paddy fields (i.e. Bajiri/Maruk, Thunessa, Wal Karabu, Weedy rice) served as favourable hosts for these nematodes. Although the severity of root knot nematode infestation observed in both types of fields is relatively mild, with a rating of approximately 2 out of 10 on the severity scale, there is a high risk of spreading the infection to the next crop. Consequently, implementing measures such as effective weed control, field sanitation, application of paddy husk charcoal, management of plant nutrients (more importantly P and K), and proper water management emerges as crucial cultural practices for managing the root-knot nematode infection in rice cultivation. These practices, in conjunction with the recommendation of utilizing resistant rice varieties, collectively contribute to a comprehensive management strategy.



- The research team with the officers from Provincial Department of Agriculture and farmers during the field inspection in Nivithigala, Rathnapura District.

References

- Bridge J, Plowright RA, Peng D. 2005. Nematode parasites of rice. *Plant parasitic nematodes in subtropical and tropical agriculture* 2: 87-130.
- Dobermann, A., & Fairhurst, T. H. (2000). Nutrient disorders and nutrient management. Potash and Phosphate Institute, Potash and Phosphate Institute of Canada and International Rice Research Institute, Singapore, 191.
- Dobermann, A., Cassman, K. G., Mamaril, C. P., & Sheehy, J. E. (1998). Management of phosphorus, potassium, and sulfur in intensive, irrigated lowland rice. *Field Crops Research*, 56(1-2), 113-138.
- Kitson, R. E., & Mellon, M. G. (1944). Colorimetric determination of phosphorus as molybdivanadophosphoric acid. *Industrial & Engineering Chemistry Analytical Edition*, 16(6), 379-383.
- Liu, M. Y., Peng, D. L., Wen, S. U., Xiang, C., Jian, J. Z., Jie, Z. H. A. O., ... & Jing, L. I. U. (2022). Potassium sulphate induces resistance of rice against the root-knot nematode *Meloidogynegraminicola*. *Journal of Integrative Agriculture*, 21(11), 3263-3277.
- Mayamulla, S., Weerathne, L. V. Y., Marambe, B., Sirisena, D. N., & Suriyagoda, L. D. B. (2017). Variation in seed nutrient content, seedling growth and yield of rice varieties grown in a paddy field without application of fertilisers for forty years. *Crop and Pasture Science*, 68(4), 337-348.
- Nelson, D. W., & Sommers, L. E. (1973). Determination of total nitrogen in plant material 1. *Agronomy journal*, 65(1), 109-112.
- Nishantha, K. M. D. W. P., Marasinghe, M. A. K. M., Wijethunga, H. M. G. D., Wijewardhane, K. G. U., Dissanayake, M. L. M. C., & Nugaliyadde, M. M. (2020). Soil Types and their Physio-Chemical Properties for Population Development of Root Knot Nematode (*Meloidogyne* spp.) in Tomato (*Solanum lycopersicum*). *European Journal of Agriculture and Food Sciences*, 2(6).
- Péret, B., Clément, M., Nussaume, L., & Desnos, T. (2011). Root developmental adaptation to phosphate starvation: better safe than sorry. *Trends in plant science*, 16(8), 442-450.
- Rose, T. J., Impa, S. M., Rose, M. T., Pariasca-Tanaka, J., Mori, A., Heuer, S., ... & Wissuwa, M. (2013). Enhancing phosphorus and zinc acquisition efficiency in rice: a critical review of root traits and their potential utility in rice breeding. *Annals of botany*, 112(2), 331-345.
- Suriyagoda, L. B. D. (2022). Rice production in nutrient-limited soils: Strategies for improving crop productivity and land sustainability. *Journal of the National Science Foundation of Sri Lanka*, 50(3).



Vandamme, E., Wissuwa, M., Rose, T., Dieng, I., Drame, K. N., Fofana, M., ... & Saito, K. (2016). Genotypic variation in grain P loading across diverse rice growing environments and implications for field P balances. *Frontiers in Plant Science*, 7, 1435.

Van Ranst, E., Verloo, M., Demeyer, A., & Pauwels J, M. *Manual for the Soil Chemistry and Fertility Laboratory-Analytical Methods for Soils and Plants, Equipment, and Management of Consumables*. NUGI 835, Ghent, Belgium (ISBN 90-76603-01-4), 1999, 243 pp.

Wang, C., Bruening, G., & Williamson, V. M. (2009). Determination of preferred pH for root-knot nematode aggregation using pluronic F-127 gel. *Journal of Chemical Ecology*, 35, 1242-1251.



- Research team with the government officers and farmers at Weeraketiya, Hambantota district

